

Question:

Carla is downloading a 200 GB file. Normally she can download 2 GB/minute, but 40% of the way through the download, Windows forces a restart to install updates, which takes 20 minutes. Then Carla has to restart the download from the beginning. How long does it take to download the file?

Reasoning + Answer:

<think> To solve this problem, let's first calculate how much of the file Carla downloads before the Windows update forces a restart. Since the file size is 200 GB and she has downloaded 40% of it before the restart, we can find this amount by calculating (200×0.40) . So, $(200 \times 0.40 = 80)$ GB are downloaded before the restart. Next, we need to account for the 20 minutes the download took to restart. Given her normal download speed of 2 GB per minute, let's calculate how much data was intended to be downloaded during those 20 minutes. We do this by multiplying her download speed by the time: $(2 \times 20 = 40)$ GB. Now, we add this 40 GB to the 80 GB she had already downloaded to find out how much she has downloaded in total before the restart: $(80 + 40 = 120)$ GB. We know the total file size is 200 GB. To find out how much data remains to be downloaded after the restart, we subtract the already downloaded amount from the total file size: $(200 - 120 = 80)$ GB. Finally, since the restart means she has to download the entire file again from the beginning, we need to add the 80 GB she still needs to download to the 120 GB she has already downloaded, which is $(120 + 80 = 200)$ GB. This confirms our final calculation. However, the question asks how long it takes her to download the file in total, considering she has already downloaded 120 GB and needs to download the remaining 80 GB from the start. To find out the total time for the new download, we calculate the time required to download the remaining 80 GB at her usual speed of 2 GB / minute. Doing the division $(80 \div 2 = 40)$ minutes for the remaining download. To find the complete total time, we add the initial 20 minutes (forced restart) to the 40 minutes it takes to download the remaining 80 GB. Thus, $(20 + 40 = 60)$ minutes. </think><answer> 60 </answer>

